

Structural Estimation of Marriage Models

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A structural approach is used to examine who matches with whom. A two-sided matching model that allows for marital sorting in response to marriage market flexibility and agents' preferences is utilized. Estimation is based on imbedding the numerical solution of a matching model within a maximum likelihood procedure. Results indicate that wage is more desirable than education in predicting marriageability for white men; education is more desirable for black men. The marriage market for white men is more flexible. Both marriage market flexibility and the chance of being classified correctly using agents' wage and education decrease with age for white men.

I. Introduction

Studies of who matches with whom have been applied to various markets such as the legal market, labor market, sports market, and marriage market.¹ How individuals sort themselves into marriage, in particular, has important implications for the distribution of income, labor supply, and fertility (Becker 1973; Boulier and Rosenzweig 1984; and Pencavel 1998).

I have benefited from suggestions of George Neumann and seminar participants in the seventh International Conference of the Society for Computational Economics at Yale University in 2001 and the Search and Matching Conference at the University of Iowa in 2000. Unfortunately, all errors are mine.

¹ For example, Spurr (1987) studies positive assortative mating between the size of law firms (legal claims) and lawyers' quality. Oi (1983) studies the matching between the number of entrepreneurs' and workers' productivity. Rosen (1981) shows that technological change leads gifted athletes to match with a larger audience. Becker (1973) proves that men and women are positively sorted in complementary traits.

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Most empirical works on who matches with whom in the marriage market have been descriptive in nature (Vandenberg 1972; Spuhler 1982; and Keller, Thiessen, and Young 1996). These studies are largely based on a reduced-form approach using spousal correlation indexes for various traits. But such an approach is ad hoc and inadequate in explaining individuals' sorting outcomes. For example, evidence shows a low spousal correlation for wages as compared with other traits such as age and education.² Does this imply that wage is not an important trait in individuals' preferences? Or does it reflect that the marriage market is inflexible, so agents scramble for partners of various wage attributes instead of being selective? Observations of who matches with whom contain more information than a simple association of spousal traits. They contain valuable information on how efficiently the marriage market functions and on agents' preferences in terms of mate choice.³ But a reduced-form approach may be unable to address market efficiency or agents' preferences that affect the value of marriage. A better understanding of who matches with whom calls for a structural investigation.

Dynamic two-sided matching models provide an appealing framework for analyzing partnership formation among heterogeneous individuals whose utility is nontransferable.⁴ They also offer a structure that characterizes the stochastic processes governing partners' arrival/separation and the choice of partners (Collins and McNamara 1990; Burdett and Coles 1997, 1999). Given these frictions, equilibrium-matching outcomes are driven by the underlying assumption of match utility/production, which affects individuals' outside options. Because two-sided matching models are relatively new, little is known about structural estimation in this context.

This essay examines who matches with whom using a structural approach based on the framework of two-sided matching models. Because the model is capable of answering how market efficiency and individuals' preferences affect who marries whom, the results give more insights into individuals' behavior than those using the simple method of spousal correlation for traits. Using the Panel Study of Income Dynamics (PSID; 1968–93), results indicate that the marriage market for white men is more flexible than that for black men. The proposed estimation method enables me to include more than one trait as individuals' types and to account

² Census evidence on spousal correlation in terms of wage, age, and education for newlyweds is provided in the appendixes.

³ Hereafter I use flexibility and efficiency interchangeably.

⁴ Dynamic two-sided models are used in this essay instead of static and frictionless two-sided models such as those of Gale and Shapley (1962) because the theory of stable matching is difficult to reconcile with facts such as the rising divorce rate. Moreover, in static models, agents' preferences are idiosyncratic, and consequently, the link between equilibrium allocation and preferences is unclear.

for classification errors.⁵ Results reject the use of one trait as individuals' overall type. More important, results indicate that inattention to classification errors leads to false predictions of marriage market inflexibility and the inaccurate conclusion that wage is undesirable. Once classification errors are accounted for, the marriage market is more flexible, and wage is found to be a more desirable measure of marriageability than education for white men. As temporal income (wage) is a more desirable trait for white men, white men are more impatient than black men, whose marriageability relies more on the permanent income of education. This result is consistent with the finding that the marriage market is more flexible for white men.

Across various age groups for white men, the flexibility of the marriage market decreases with age, and the chance of being classified incorrectly increases with age. However, the effects of age for black men are mixed. Despite the dissimilarity in marriage behavior between blacks and whites, both groups exhibit a U-shaped "age-wage" profile, that is, wage is least important as a measure of marriageability around the middle-aged group.

Empirical results are based on a two-sided matching framework in the spirit of Burdett and Coles (1997) and Collins and McNamara (1990), where individuals' match utility is nontransferable and matching is random. In the model, agents are *ex ante* heterogeneous, with each represented by a marriage index. Finding potential matches is a time-consuming process, characterized by scarcity of information. Given such a costly matching process generated by search friction, a range of acceptable potential partners is possible. Agents choose an acceptable set of partners to maximize expected wealth. The equilibrium of who matches with whom is affected by the structural parameters that characterize the distribution of the marriage index and the frequency of agents' marriage market transitions.

Estimation of the structural parameters of a matching model is done where observed couples are considered as an equilibrium outcome. The estimation method developed in this essay improves on empirical works describing who matches with whom in the following ways. First, the method of ranking agents based on multiple measured attributes such as wage and education is more general than using a single trait as agents' types in standard literature (Boulier and Rosenzweig 1984; Montgomery and Sulak 1989). Second, a flexible classification error model is developed. Third, the method of estimation is straightforward. Given a distribution of agents' marriage indexes and the frequency of marriage market contacts and separations, the acceptable pool of partners for each type of agent can be solved numerically following an optimal reservation-match policy.

⁵ These two measures are less likely to be implementable in a simple correlation index.

The numerical solution is then nested within a maximum likelihood procedure to estimate structural parameters. The estimation technique is applied to samples of the PSID. Since the probability of who matches with whom and the spell length of looking for partners are endogenously determined by the optimal strategy, I show that observations of the age at first marriage and the characteristics of matched partners provide information on identifying structural parameters.

Fourth, individual marriage history is used rather than cross-section data, which record a stock of individuals at a given point in time. The advantage of using marriage history data is twofold. Agents' marriageable traits are drawn at the time of marriage because a pool of single agents is followed until they get married. This avoids the problems in most studies that use postmarriage characteristics from cross-section data.⁶ These problems include division of labor within a family that may affect choices of occupation and the labor supply, so postmarriage wage may be correlated with marital choice. Another advantage of using panel data is that the structure of data allows a direct application to the dynamic nature of the model, thereby allowing direct testing of the model's assumptions and, hence, testing of its usefulness.

This essay is similar to Collins and McNamara (1990); Burdett and Coles (1997, 1999), and Bloch and Ryder (1998), which study partnership formation among heterogeneous agents. But unlike those and as in Smith (1997), this essay utilizes productive aspects of marriage. In this essay, agents are productive while single and while married. Further, unlike Bloch and Ryder (1998), who assume cloning, I assume exogenous separation to achieve steady state singlehood distribution. I choose exogenous separation because it is more realistic than cloning, and it gets rid of the second infinity problem.⁷

This essay is similar in technology to Mortensen and Neumann (1988) in the sense that the reservation function is computed as a fixed point of a contraction map. Mortensen and Neumann perform Monte Carlo evidence on the properties of structural estimation of a homogeneous search model when solving for exact reservation wage. This essay is also similar in two ways to typical applications in search models and equilibrium search models such as those of Bunzel et al. (1997) and Bontemp, Robin, and van den Berg (1999). First, these studies consider steady state reservation solution and the estimation of structural parameters in a continuous time and infinite horizon model. Second, these applications take

⁶ See, e.g., Boulier and Rosenzweig (1984); Montgomery and Sulak (1989); Bergstrom and Schoeni (1996); and Suen and Lui (1999).

⁷ This assumption is also less complicated than exogenous inflow and endogenous entry. Endogenous entry is used in standard search-matching literature in the labor market to determine the number of vacancies (Pissarides 1990).

workers' labor market history and the accepted wage at employment to identify the intensity parameters in the stochastic processes and to determine the endogenous productivity distribution or wage distribution. In this essay, identification requires knowledge of agents' marriage history and their characteristics at the time of the formation of matches. Such formulation allows one to test the restrictive assumptions on the memoryless properties of partners' arrival and the separation rates made in the model.

The matching model is outlined in Section II. Section III contains a discussion of the empirical implementation of the model. The structural model is estimated by maximum likelihood. I derive the likelihood function of the matching model in Section IIIA, and in Section IIIB, I propose a discrete index as an attempt to rank agents' measured attributes. I present a matching algorithm in Section IIIC. I close the section by deriving estimation strategies that take into account classification errors in estimation. Section IV contains a description of the data. Estimation results are given in Section V. I present estimates of partners' arrival rate, separation rate, and wage elasticity for distinct age categories. Section VI concludes.

II. The Matching Model

Time is continuous, and there are two groups of infinite-lived agents: men and women. At each point in time, all agents are in one of two states: single or married. Only single agents search for marriage partners. Let the arrival rate of partners for single agents be λ , which is governed by a Poisson process. Given an arrival of partners, agents decide whether to form a match. If a match is formed, it dissolves exogenously at rate δ , and agents flow back to the single pool. If no match is formed, then an individual searches again. Thus, the market is described by the structural parameters that describe market transitions $\langle \lambda, \delta \rangle$.

Agents are ex ante heterogeneous with respect to their types x_i . Let there be J types for men and women, where J is a positive finite integer bounded away from ∞ . The range of types is $[\underline{x}_i, \bar{x}_i]$, where $\underline{x}_i$ and $\bar{x}_i$ indicate the infimum and supremum of its support. Assume for illustration that the distribution of types is the same for men and women.⁸ Let γ^i denote the distribution of types among single men who will propose to a type j woman if they meet, that is, $\gamma^i = \sum_{k=1}^J \gamma_k = \Pr(X < x_{i+1} | x_j)$, where γ_j denotes the corresponding probability density function. Types are distributed according to γ^i , independently of spells of being single.

While an agent is single, the instantaneous utility is the real value of the agent's type. When an agent is married, instantaneous utility is assumed to be an equal split of match production $x_i x_j$, where i represents

⁸ This assumption will be relaxed in the empirical estimation.

the type of men and j represents the type of women. The utility structure implies that $\underline{x}_i \geq 2$.⁹ The types of two potential partners are revealed upon meeting, and each agent must decide whether or not to accept a proposal.

A type i agent chooses a range of acceptable types of potential partners with the objective of maximizing his expected discounted value in the future utility stream. A single agent has instantaneous utility x_i and the expected benefit of marriage following an optimal policy if partner type Y is realized, given that a partner has arrived. If agents discount future income at rate β , the value of being single is

$$V(x_i) = \frac{x_i + \lambda E \max [W(x_i, Y), V(x_i)]}{(\beta + \lambda)}, \quad (1)$$

where $W(x_i, Y)$ is the expected discounted value of marriage with a random partner of type Y . The ex post value of marriage is made up of the match utility given by the equal split of the realized match production and the value of remaining single due to an exponential random separation,

$$W(x_i, x_j) = \frac{x_i x_j}{2(\beta + \delta)} + \frac{\delta V(x_i)}{(\beta + \delta)}. \quad (2)$$

A marriage proposal is acceptable if the potential partner is above the value of the reservation type R_i of type i . Let r represent the ranking of R , so $R_i = x_r$. Not all potential partners whose type is above r accept marriage proposals offered by given agents in equilibrium. Some potential partners are simply unattainable because they may prefer not to match with lower-type agents. This feature is the equilibrium outcome of the model and not a restriction. Denote M_i to be the value of the maximum attainable type and m to be the ranking of M , so $M_i = x_m$. The acceptance set of a type i agent is $A_i = \{j | r_i \leq j \leq m_i\}$. The probability of acceptance is $\Pr(A_i) = \gamma^{m_i} - \gamma^{r_i}$.

The optimal policy is a reservation-match policy given by $W(x_i, R_i) = V(x_i)$. If a marriage offer falls within the agent's acceptance set, the agent will accept the match proposal following the optimal policy; otherwise, the offer will be rejected. Combining equations (1), (2), and the optimal policy, the reservation type is the solution to the following equation (see app. A for derivation):

$$R_i = 2 + \frac{\lambda}{\beta + \delta} \sum_{j \in A_i} (x_j - R_i) \gamma_j. \quad (3)$$

⁹ The infimum $\underline{x}_i$ must be at least as large as two to satisfy the incentive constraint for marriage: that the match utility is at least as large as the utility while single, $x_i x_j / 2 \geq x_i (x_i x_j / 2 \geq x_i)$. Should this constraint fail to hold, a fraction of each sex will not be married.

The solution is unique because the left-hand side of equation (3) is increasing in R_i , and the right-hand side is decreasing. Since the situation is symmetric between men and women, the reservation type for women satisfies

$$R_j = 2 + \frac{\lambda}{\beta + \delta} \sum_{i \in A_j} (x_i - R_j) \gamma_i. \quad (4)$$

A Nash equilibrium of matching requires that (i) each agent find a potential partner acceptable, and (ii) every single agent select his (her) own partner type to maximize the expected net benefit flow attributable to the choice of partner following equation (3) (or eq. [4]), given the optimal choices made by all other single agents. In other words, two sets of joint matching strategies for men and women, $\{R_i, M_i\}$, $\{R_j, M_j\}$, for all $i, j = 1, \dots, J$, that solve equations (3) and (4) are Nash equilibrium solutions to the noncooperative stationary game of matching.¹⁰ The equilibrium solution is a two-dimensional graph.

Positive assortative matching, that is, a positive relation between reservation types and agents' own types, is the predicted equilibrium outcome. This result is driven by the underlying specifications of agents' output, which is agents' type when single and the product of each partner's type when married. The match output encompasses increasing returns to scale as an important gain from marriage. This property is necessary to create incentives to trade with higher-type agents to exploit match rents when narcissism is allowed. When utility of being single is agents' own types, waiting for higher-type partners is less costly, and agents who were previously accepting lower-type partners will prefer to wait for higher-type partners. Agents' outside options must be sufficiently high to induce positive assortative matching. If a constant returns to scale match output such as $\sqrt{x_i x_j}$ or $x_i + x_j$ is assumed, negative assortative matching arises.¹¹ When agents do not value their singlehood utility, or when search friction is absent, increasing returns to scale is not necessary to induce positive assortative matching.¹²

In this model an increase in reservation type can be explained by either a rise in the arrival rate of partners, a fall in the separation rate, or a combination of both. The ratio $\lambda/r + \delta$ can be thought of as a measure of the inverse of search friction. This is a ratio of partners' arrival rate to the sum of the discount rate and separation rate, and so it measures the relative speed of offers. As $\lambda \rightarrow \infty$, partners arrive instantaneously. Friction

¹⁰ For a proof of the existence of the equilibrium, see Smith (1997).

¹¹ See Wong (1998) for discussions.

¹² Burdett and Coles (1997, 1999) characterize positive assortative matching when narcissism is ruled out. Under a frictionless framework, Becker (1973) shows that monotonic preferences are sufficient for positive assortative matching.

disappears, each type of agent matches with his or her most desirable type, and strict positive assortative matching results, as in a centralized market. When chances to meet partners are low, $\lambda \rightarrow 0$, all agents are least selective. For values of λ between zero and infinity, a range of partner types is acceptable.

III. Estimation Strategy

The object of interest is to estimate the likelihood of a type i agent marrying a type j agent. The method used is maximum likelihood, for two reasons. First, the outcome of the matching model endogenously generates an acceptance set for each type of agent, which allows the derivation of the probability of who matches with whom. Second, the assumptions underlying the model enable derivations of the distributions of duration data.

A. The Likelihood Function

Identification of all parameters requires knowledge of couples' characteristics at first marriage as well as an individual's marriage history. All agents are single at the first interview, whence information on how long they have been single can be obtained. Information on the duration of singlehood is obtained by following single agents after the first interview. Analogous to the renewal theory, these durations are backward T_{ob} and forward T_{of} recurrence times, respectively. Therefore, the duration of singlehood is $T_0 = T_{ob} + T_{of}$. The data consist of a panel where some individuals are single with duration T_0 , some are married with duration T_1 , and a given individual's and spouse's logarithms of wage and education at first marriage are w and e , respectively. Assume that the parameters of males and females are the same; the structural parameters to be estimated are λ and δ .

Consider a type i man who is single at first interview. Let T_{ob} and T_{of} be independently and identically distributed and have an exponential distribution with parameter $\lambda(\gamma^m - \gamma^r)$. Let D_{ob} (D_{of}) denote a binary variable that equals one if it is known that the elapsed (residual) duration exceeds a certain value, that is, left-censored (right-censored), and zero otherwise. Conditioned on being type i , the individual contribution of singlehood duration until and including the time of exit into marriage or censoring is

$$L_{oi} = [\lambda(\gamma^m - \gamma^r)]^{1-D_{ob}+1-D_{of}} \exp[-\lambda(\gamma^m - \gamma^r)(T_{ob} + T_{of})], \quad (5)$$

where $T_{of} > 0$, and $T_{ob} > 0$.

Events occurring after exit from being singlehood are independent of the events up to exit. Therefore, their probability is independent of the likelihood of being singlehood. The event immediately following type i 's

singlehood duration is the realization of with whom to match. This event is given by the density of accepted type, $f(x_j|x_i) = f(x_j|j \in A_i)$. Let N_j be the number of type j agents, and $\sum_j N_j I(j \in A_i)$ be the number of potential partners acceptable to a type i man, where $I(j \in A_i)$ is an indicator function equaling one if a type j woman is acceptable to a type i man. The acceptance criterion of a type i man is endogenously determined by solving equations (3) and (4). Given a type i man, the probability that the type i man matches with a type j woman is the number of type j women out of all types of women acceptable to a type i man,

$$f(x_j|x_i) = \frac{N_j I(j \in A_i)}{\sum_{j=1}^J N_j I(j \in A_i)}. \quad (6)$$

Conditional on the realized partner type, marriage duration T_1 has an exponential distribution with parameter δ . If $D_{of} = 1$, I do not follow the individual any longer. Let D_2 equal one if T_1 is right-censored, and equal zero otherwise. If $D_{of} = 0$, then a type i individual's likelihood contribution to events between entering marriage and separation equals

$$L_{1ij} = f(x_j|x_i) \delta^{1-D_2} \exp(-\delta T_1), \quad (7)$$

where $T_1 > 0$. The total type i individual likelihood contribution for a respondent who is single at the time of the first interview equals the product of equations (5) and (7) that describes the odds of each type i man who initially is single matching with a type j partner with a marriage offer, followed by the marriage dissolving exogenously:

$$L_{ij} = L_{0i} L_{1ij}^{(1-D_{of})}. \quad (8)$$

Since observations of each type of men are independent, if n denotes the n th observation of men, the likelihood function of the benchmark model is

$$L_b = \prod_{n \in (i,j)} L_{ij}, \quad (9)$$

where $i = 1, \dots, J$, and $j = 1, 2, \dots, J$.

Before launching the likelihood estimation, two operations must be performed. The first is to rank an individual's type based on observed data. The second is to obtain the analytical acceptance set for each type of individual based on the model. Therefore, the likelihood function must nest within it a matching algorithm to solve for the acceptance set of each type of individual based on equations (3) and (4).

B. The Marriage Index

Types are marriage indexes of individuals. To estimate the model in a manageable way, I assume types are generated by two measured attributes:

the individual's logarithm of wage w and education e .¹³ Then I rank individuals into discrete categories according to the following steps:¹⁴

- a) Generate z as a function of w and e

$$z = \exp[\alpha w + (1 - \alpha)e], \quad (10)$$

where α is a scalar parameter. The α measures the sensitivity of spousal demand of a change in wages. If spousal demand is sensitive to a change in wage, then wage reflects a more important marriageable characteristic relative to education.

- b) Take the range of the corresponding order statistics of z and discretize it into J equal partitions, where J is a positive finite integer. For example, if $J = 10$ and the sample size is N , $z \leq \dots \leq z_{(N)}$ is partitioned into the decile. Within the i th interval, z is bounded by $z_{Li} < z \leq z_{Hi}$, where z_{Li} indicates the lowest z that makes a type i individual, and z_{Hi} indicates the highest z that makes a type i individual.
- c) The set of z within each i th interval is mapped to x_i , following the rule:

$$x_i = \text{median}[z_{Li} < z \leq z_{Hi}]. \quad (11)$$

The real-valued x_i represents a type i individual, which is a piecewise constant within the corresponding i th interval of $z_{(1)} \leq \dots \leq z_{(N)}$. Thus, given α , w , and e , $x_i(\alpha)$, $x_j(\alpha)$, and the corresponding empirical type distributions are generated.

C. The Matching Algorithm

Equilibrium matching sets are endogenously determined in the model. All parameters $(\lambda, \delta, \alpha)$ are assumed to be the same between men and women, but type distributions need not be the same because of gender difference in observed attributes. Given a vector of parameter values $(\lambda, \delta, \alpha)$, a matching algorithm is proposed to compute the acceptable pool of partners for each type of individual for the matching model. The difficulty in solving for the reservation type and the highest attainable type for each type of individual involves constantly updating the acceptance criteria from other individuals. The acceptable pool of partners can be identified by solving the equilibrium acceptance set backward, starting from the highest type from each gender side.

The Matching Algorithm

Step 1. For the highest type of men and women, set the maximum attainable type to J .

¹³ Classification errors are introduced in Sec. IIID.

¹⁴ Since the procedures of constructing "type" to each sex is the same, I drop the gender subscript.

Step 2. Use equations (3) and (4) to solve for R for the highest type of men and women, respectively. $\{R_{i=j}, M_{i=j}\} \times \{R_{j=j}, M_{j=j}\}$ defines the first acceptance area.

Step 3. For any i th type individuals, where $i < J$, two cases can occur:

Case 1. M_i is identifiable. This occurs when $\{j | R_j \leq x_i\}$ is not empty for some $j > j'$. In this case, i is accepted by some $j > j'$. We set $M_i = \max_{j > j'} \{j | R_j \leq x_i\}$ and solve for R_i . Repeat for the women's side.

Case 2. M_i is unidentifiable. This occurs when $\{j | R_j \leq x_i\}$ is empty for a $j > j'$. In this case, we reverse the role of i and j and solve for $\{R_j, M_j\}$, aiming at determining the acceptance sets of additional $j = j - 1, j - 2, \dots$, until the first woman accepts the i th man, that is, $R_j \leq x_i$. However, to solve for $\{R_j, M_j\}$, we need to be able to determine $M_j = \max_{i > i'} \{i | R_i \leq x_j\}$, which may not be possible if the set is empty. If we hit an empty set of $\{i | R_i \leq x_j\}$ before obtaining $R_j \leq x_i$, reset M_j to x_i and M_i to x_j at the point of failure, and repeat step 3 again.

Stopping Criterion

We have solved for all acceptable sets of partners for type i and type j . If there are remaining types of i or j that have not been matched, they will be assigned a null acceptance area (no matching possible).

D. Classification Errors

Just as search models are sensitive to measurement errors, the matching model is sensitive to classification errors. To resolve the problem, I develop a classification error model.¹⁵

The classification error model proposed here is simple and flexible. Let k and l denote the true type for i and j , respectively. Let the classification errors for type i and j agents, v_1 and v_2 , be independently and identically distributed, where $i = k + v_1$ and $j = l + v_2$. Because the supports of v_1 and v_2 are known, only their distributions need to be estimated.

Let q_{ik} be the probability of classification error for a type i agent, and q_{jl} be the probability of classification error for a type j agent. Because classification is assumed to be exhaustive and mutually exclusive, the

¹⁵ From a technical standpoint, a classification error model is needed. The reason is that eq. (6) indicates that estimates must accommodate the selection constraint $I(j \in A_i) = 0$; otherwise, the contribution to the likelihood function in eq. (9) is zero. Imposing such a condition distorts parameter estimates, and estimation from the model may yield misleading conclusions. Consider a marriage market in which most people match with partners of similar types, except for a few serious outliers, e.g., the highest type matches with the lowest type. Then, parameter estimates must accommodate the matching of extreme types, and the matching model implies that a sufficiently low partners' arrival rate λ is necessary to sustain all couplings. Thus, instead of obtaining estimates that support the sorting of partners of similar types, estimation results would yield estimates that produce acceptance probability equal to one, search friction would be erroneously high, and assortative matching would be erroneously low.

sum of the probabilities for a given observed category is one, that is, $\sum_k q_{ik} = 1$. Let $\mathbf{Q}$ denote a $J \times J$ matrix of the probability of all possible classification errors, with elements q_{ik} . The exhaustive assumption forms a total of J restrictions on possible values for q_i . Thus, $\mathbf{Q}$ contains a total of $J(J-1)$ free parameters. Suppose there are 10 types of agents. The total parameters of classification errors for both men and women are 180.

To reduce the parameter space, I consider the problem of classification errors in terms of the distance between true and observed type in the following way. Let $d(k, i) = |k - i| = |v_1|$ be the distance between the true type k and the observed type i , and $q(d)$ denotes the classification error with distance equals to d . I make two assumptions.

Assumption 1. Symmetry: the probability of misclassifying an individual is the same for any i and k with the same distance, $q(d) = q(d')$ for any $|k - i| = |k' - i'|$.

Assumption 2. The error distributions are the same for both sexes and are independent of sex.¹⁶

The number of types is set to 10, that is, $J = 10$. The following tabulation represents $\mathbf{Q}$, in which the columns represent observed type i , the rows represent true type k , and the entries represent $q(d)$:

	Type i									
	1	2	3	4	5	6	7	8	9	10
1	$q(0)$	$q(1)$	$q(2)$	$q(3)$	$q(4)$	$q(5)$	$q(6)$	$q(7)$	$q(8)$	$q(9)$
2	$q(1)$	$q(0)$	$q(1)$	$q(2)$	$q(3)$					
3	$q(2)$	$q(1)$	$q(0)$							
4	$q(3)$	$q(2)$		.						
5	$q(4)$	$q(3)$			.					
6	$q(5)$					.				
7	$q(6)$						.			
8	$q(7)$							.		
9	$q(8)$								.	
10	$q(9)$									$q(0)$

If types are observed without errors, $\mathbf{Q}$ is a diagonal matrix with elements equal to one, that is, $q(0) = 1$, and $q(d) = 0$, for $d > 0$. Since there are at most 10 errors, $q(0), q(1), \dots, q(9)$, the parameter space is reduced to nine. In general, this restriction reduces the number of free parameters from $J(J-1)$ to $(J-1)$. Note that the exhaustive condition gives rise to the following equality:

$$\mathbf{A}\mathbf{q} = \mathbf{1}, \quad (12)$$

where $\mathbf{A}$ represents a 10×10 matrix containing the number of occurrences of error probabilities associated with each element of a 10×1 vector $\mathbf{q} = [q(0), q(1), \dots, q(9)]^T$, and $\mathbf{1}$ is a 10×1 unit vector. In other

¹⁶ The error structure is assumed to be symmetric between men and women, and so parameters of the classification errors are identical for men and women. The forthcoming discussions on q_{ik} apply to q_{ji} .

words, equation (12) can be written as

$$\mathbf{A}\mathbf{q} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 2 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 2 & 2 & 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 2 & 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 2 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \times \begin{pmatrix} q(0) \\ q(1) \\ q(2) \\ q(3) \\ q(4) \\ q(5) \\ q(6) \\ q(7) \\ q(8) \\ q(9) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Clearly, rank $\mathbf{A}$ is $5 < J = 10$. The solution to $\mathbf{q}$ is

$$\mathbf{q} = \begin{pmatrix} 1 - 2e_1 - 2e_2 - 2e_3 - 2e_4 - e_5 \\ e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_4 \\ e_3 \\ e_2 \\ e_1 \end{pmatrix}. \quad (13)$$

However, not any solution of $e_1, \dots, e_5$ is admissible. The solutions for $e_1, \dots, e_5$ have to be chosen such that the following two conditions must be satisfied: (a) $e_1, e_2, e_3, e_4, e_5 > 0$, and (b) $2e_1 + 2e_2 + 2e_3 + 2e_4 + e_5 \leq 1$.

For all sample of men, the likelihood function, satisfying equation (13a, b) becomes

$$L_c = \prod_{n \in (i,j)} \sum_{v_1(e) = i-10}^{i-1} \sum_{v_2(e) = j-10}^{j-1} L_{\mathcal{Q}(i-v_1(e))} [L_{1(i-v_1(e))(j-v_2(e))} q(|v_2(e)|)]^{(1-D_{\mathcal{Q}})} q(|v_1(e)|), \quad (14)$$

where $i = 1, \dots, 10$, and $j = ., 1, 2, \dots, 10$.

IV. Data

A. The Sample

To estimate the matching model, panel data with marriage history are used. I consider individuals who transited from being single to married between 1968 and 1993 and follow their marriage histories thereafter. Specifically, I use data on the age at first marriage, the couple's wages and

education at first marriage, and the duration of marriage. Although longer marriage histories are available, I use only the first two spells.

I based my findings on PSID 1968–93 family and individual samples.¹⁷ The family files do not contain the marriage history of the respondents, and the individual file does not contain detailed demographic and employment data. I first use the individual file to create an eligible sample population with well-defined marriage history, and then link the sample to family files to obtain for each household head, and spouse if married, their corresponding demographic and employment data.

The 1968–93 individuals file contains 53,013 respondents ever interviewed with their corresponding marriage histories, including information on when first marriages started and ended (if this happened). I exclude individuals with inconsistent marriage histories, for example, those with uncertain first marriage years or marriage termination years, or those who ended the first marriage before their first marriage started. Imposing this restriction, only 27,741 respondents are left.

I keep samples of those respondents whose first marriages began in or after 1968 and lasted until or after 1985. For those individuals whose first marriages began before 1968, there was no information on the first age at marriage or the corresponding marriageable characteristics. The race of household wives were not available in the PSID before 1985. Thus, for couples whose first marriages dissolved before 1985, there was no race information. This selection leaves me with 20,441 observations. The PSID oversamples Hispanics. After excluding the Latino sample, 15,193 observations are left.

A substantial fraction of the original sample is not present after 25 years of survey.¹⁸ Attrition occurred as a result of loss of contact as families moved, maturing children could not be traced, or respondents refused to continue to be interviewed. Observations are not used for single respondents if their marriage histories were last updated before 1993. For married couples, observations are not used when marriages were not updated in 1993 but first marriages lasted after 1993. Imposing this restriction, the

¹⁷ Respondents have been interviewed annually since 1968. Although the PSID also has data for 1994–96 in early release versions, an individual file cannot be accessed in a cross-year way as in the 1968–93 file, and thus individuals from year to year cannot be followed.

¹⁸ By 1993, only 122 out of 1,181 households came from the 1968 sample, 20% of which were married. Due to the small sample size, I chose the 1968–93 sample, which included those born after 1968 who became age 15 or older in 1993. Duncan and Hill (1989) documents the representativeness of the PSID (1968) after 17 years. They find that there is a serious problem of attrition, and most of the original households are not represented by respondents in 1968. However, the samples still have mean characteristics comparable to those in 1968.

sample population is reduced to 11,141. This sample contains individuals from the main family and the subfamily of each household.

To create data linking members of a couple, I use information on the household head and spouse (or cohabitor) because each family file contains demographic and employment data only for household heads (and spouses or cohabitators), not respondents from the subfamilies. Thus, I further restrict the marriage sample so that both household heads and spouses (or cohabitators) were present. In addition, only one-fifth of the sample contains the marriage histories of single female heads who transited to marriage, and so I focus on men's marriage histories. This leaves a total of 2,855 households, of which 1,517 are single individuals, and 1,338 are married couples.

Search duration can only be partially observed because the elapsed singlehood duration T_{ob} is unknown.¹⁹ In what follows, I use age 15 as an index for spousal search starting time.²⁰ Therefore, if T^* is the stopping time of being single, and C is the censoring time, the completed spell of search duration is $T_o = T_{ob} + T_{of} = \min\{T^*, C\} - 15$. The duration of marriage is defined as the number of years a couple stays married before or until the censored time, whichever comes first. I do not use samples with uncertain race and age records, nor when respondents have ever been institutionalized. This leaves a total of 633 married couples and 1,264 single individuals.

Data concerning wages and education of respondents (and their spouses if married) are taken as of the year of their marriages.²¹ I assume that these characteristics are time invariant. I cross-check all time and pay rate responses²² against the upper and lower bounds collected for men and

¹⁹ The initial condition problem is solved by Chamberlain (1979) using a Bayesian technique, in which the random effect distribution is conditioned on forward recurrence information. Ondrich (1985) controls for heterogeneity, assuming that both unemployment and employment spells have Weibull distribution with a parametric unobserved heterogeneity. Results from an exponential model and Cox's proportional hazard model reveal that there is a significant heterogeneity in the duration of being single in my sample. Heterogeneity in my model is captured by the acceptance selection of each individual type, assuming that $\lambda > 0$.

²⁰ Age 15 is used because it is the official census definition for the marriageable age (see *Statistical Abstracts 1996* [U.S. Department of Commerce 1996] for details). Moreover, my sample does not contain respondents who married at ages younger than 15, and only two cases of zero single duration spells, so the choice of 15 as the starting age does not seem unreasonable.

²¹ I decode the interval data of education in 1968–74 and 1985–90, using auxiliary relations with the 1980 Annual Demographic March Current Population Survey (CPS) data.

²² I use the fifth and the ninety-fifth percentiles from the set of hourly wages for paid hourly workers from the March outgoing rotation groups for each year.

women of the same education from the CPS for the period.²³ Those whose wages do not fall within the admissible ranges are treated as missing. This gives me the final total of 443 married couples and 696 single individuals.

B. Data Description

There are five race groups in the sample: whites, blacks, Chinese, Japanese, and American Indians. Due to the small observation of some groups, I focus on white and black men only.²⁴ Table B1 in appendix B contains sample characteristics on average durations of being single (uncensored spells only), average wages and education at first marriage, average durations of marriage (including censored spells), and the proportion of complete singlehood spells and marriage spells stratified by race. The number of observations (N) corresponds to uncensored single data, in which all individuals transit into marriage.

The data reveal that, on average, white men take a shorter time (1.11 years) to search for spouses than black men. The duration of marriages for white men is longer. The proportion of the complete (or interrupted) marriage spell for white men is 11.2%, which is 3.7% higher than for black men. Wage and education data tell the conventional story: blacks earn less and have lower education levels than whites. Matchings of whites are more assortative in wage and education than those of blacks.²⁵

V. Results

Estimation results are based on categorizing agents into 10 types, that is, $J = 10$, and setting $\beta = 0.05$. My estimation strategy is applied to three specifications of α in order to separate out the contributions of wage and education in explaining who matches with whom: $\alpha = 1, 0$, and $(0, 1)$. Tables B2 and B3 show the estimated results for white and black men, respectively.

When wage is used as the marriage index, white men are about 4.5

²³ Because 27% of married women in the sample did not work, I compute potential wages for them using Heckman's two-step procedure. To correct for the selectivity bias, I estimate a participation probit using the standard Heckman procedure. The probit equation contains all variables in the wage equation and the number of children. The wage equation is controlled for by husbands' and wives' ages, ages squared, education, region dummies, city populations, and husbands' wages. The wage equation and the probit equation are estimated separately by race.

²⁴ Whites and blacks represent over 98.25% of the sample population for men and over 98.42% of the sample population for women. However, I use the entire sample for the purpose of generating the marriage index distribution, which depends on the characteristics of the sample population.

²⁵ Logarithms of wages are used to compute the wage correlation to adjust for skewedness. When the imputed wage for women is not used, the correlation is 0.472 and 0.376 for white men and black men, respectively.

times more likely than black men to contact partners.²⁶ The separation rate is also higher for white men. The marriage market for white men is more than four times as flexible (24.24 vs. 5.87) as that for black men, indicating more assortative matching for white men. This result is consistent with data showing that white men had higher correlations in wage with their spouses than black men. When using education as the marriage index, white and black men exhibit similar stratification patterns because of similar search efficiency.

Once classification error is accounted for, matching becomes more sorted.²⁷ Moreover, the marriage market is more flexible when using wage (as opposed to education) as marriageability. For example, white men's marriage market shows that $\lambda = 2.4$ when $\alpha = 0$ and $\lambda = 14.23$ when $\alpha = 1$. This result contrasts with the simple wage correlation in Section IVB, which suggests spouses are sorted similarly in wage and education. This result and the significance of classification error probabilities indicate the empirical importance of accounting for classification errors.

When using both wage and education, a significant racial contrast in the relative importance of wage occurs: 0.816 and 0.031 for white men and black men, respectively.²⁸ This indicates that wage has a greater impact on white men's desirability as marriage partners, whereas education constitutes almost all marriageability for black men. Alternatively, as wage represents temporal income, and education represents permanent income, α can be interpreted as the time rate of preference. The high α of white men implies that white men are more impatient than black men, which is consistent with the result of a higher $\lambda/\beta + \delta$ that leads to a more

²⁶ I am aware of the systematic difference in wage during the life cycle. I reestimate the model using standardized wage (by mean age), and I also use the present value of expected income as agents' trait. I find that there is no qualitative difference in results.

²⁷ Results from a benchmark model (without classification errors) show that in all α specifications, black men have lower partners' arrival and separation rates than white men. The estimates of λ for black and white men are 0.039 and 0.034, respectively, when $\alpha = 1$, and 0.032 and 0.027, respectively, when $\alpha = 0$. So the acceptance sets for both race groups range from the lowest to the highest types. That is to say, sample individuals scramble for partners, and the acceptance probability is one. The values of market efficiency are found to be close to zero, indicating that search friction is high.

²⁸ Results on the estimates of α may be affected by using imputed wages for nonworking women (27% in the sample). Because people may self-select to not working, assigning them positive wages leads to an overestimation of α . So, the estimated explanatory power of wage may be too large. Even if only agents of positive wages are included, overestimation of α would still occur, and wage as a marriageable trait tends to predict more assortative mating. However, results from Monte Carlo simulations indicate that α is underestimated. So, the net effect of over- and underestimation of α may cancel out.

efficient market for white men.²⁹ Likelihood ratio tests reject specifications using only one trait.

To examine whether there exists heterogeneity of search behavior across age groups, I stratify the sample by four age groups: 15–21, 22–30, 31–40, and 41–65. Within each age group, type described by both wage and education is significantly different from models when only wage or education is used. Besides, the marriage market for white men is most efficient when using both traits, while the picture for black men is less clear.

When examining across age groups for $\alpha \in (0, 1)$, wage demonstrates itself as a better representation of white men's marriageability than education, except for the age group 31–40. In contrast, education is a more desirable marriageable trait for black men except the teens. Nonetheless, both blacks and whites exhibit a U-shaped "age-wage" profile.

Interestingly, there is a striking racial contrast in results from the effects of classification errors and market efficiency across age groups. Results for white men show that the chance of being classified incorrectly increases with age. This indicates that as agents get more seasoned, observable characteristics in wage and education do not reflect marriageability as well. Further, the marriage market of younger white men is more efficient than that for older agents (except for $\alpha = 0$ in table B2). However, there is no tractable effect of age for black men. Given a particular specification of α , both e_o and $\lambda/(\beta + \delta)$ may increase or decrease with age. For example, using wage only as a marriage index reveals little about older black men's marriageability (e_o is low), but it reveals considerable information about young black agents' marriageability. As education or a combination of wage and education is used as a marriage index, black men of age groups 22–30 and 31–40 are more prone to classification errors. A mixture of results is also found for the marriage efficiency of black men. The market is more efficient at both ends of the age distribution when education is the marriageable trait. But when wage only is used, the market is more efficient for age groups 22–30 and 31–40. At $\alpha \in (0, 1)$, younger black men have a more efficient market; this result resembles that of white men.

On comparing the sum of the maximum log likelihoods by age groups to the maximum log likelihood using the restricted model in table B2 (under "indexed by wage and education"), the hypothesis of homogeneity

²⁹ Comparisons of these results to those from the benchmark model (without classification errors) reveal significant shifts of the estimate of α : from 0.005 to 0.807 for white men and from 0.005 to 0.031 for black men. The estimates of λ become higher, but those of δ become smaller in the classification error model. The opposite movements of the estimates of λ and δ allow more assortative matching as described in Sec. II. When classification errors are incorporated in the model, the transition parameters adjust to allow for more assortative matching.

across subsamples is clearly rejected. This result is found in all α specifications for white and black men.

The estimated acceptance sets for white and black men are shown in figures C1–C6 of appendix C. The data are generated following equation (11) and are based on the estimates in tables B2 and B3 using the overall data. Disjoint marriage classes in the spirit of Burdett and Coles (1997, 1999) are obtained. When wage is the only determinant of the marriage index, there are three marriage classes, and they are asymmetric (fig. C1). The asymmetry is due to the substantial difference in wage distribution between white men and women. White men's types who are type 5 or above account for more than 5% from the sample, while it is about 3% for women. Because there are more high-type white men than high-type women, high-type women are more picky than men. Women of type 5 through type 10 have acceptance sets $A_{j=5} = \{i | i = 5, \dots, 10\} = A_{j=6} = \dots = A_{j=10}$, while white men of type 5 through type 10 have acceptance sets $A_{i=5} = \{j | j = 4, \dots, 10\} = A_{i=6} = \dots = A_{i=10}$. These acceptance sets form the first marriage class. The second marriage class is made up of type 2 and type 3 women and type 3 and type 4 men. The last marriage class is made up of type 1 women and type 2 men. Because women are picky, type 1 men remain single in this equilibrium.

Because search friction is highest when education is used as the marriage index, white men are the least sorted: type 3 through type 10 agents share the same acceptance set, and type 1 and type 2 agents are sorted with their own type (fig. C2). When both wage and education are considered, there are four marriage classes for white men. Men and women of type 4 through type 10 are in the same class, and strict sorting occurs in the rest of the three types (fig. C3). Because the market is most flexible in this case, agents are most sorted.

Given a higher search friction, the marriage market for black men is less sorted. In all α specifications, similar offer arrivals lead to same acceptance sets of black men, with three marriage classes in each model (figs. C4–C6).

The matching model posts strong assumptions on the distribution of singlehood and marriage spells: both follow exponential distributions with intensity parameters $\lambda(\gamma^m - \gamma^r)$ and δ , respectively.³⁰ To check how well the exponential model fits the data, I perform a formal test by fitting a Weibull model to the duration data; the purpose is to test the slope of the shape parameter ρ in the Weibull model. Under the null hypothesis of an exponential model, $\rho = 1$.

Results are shown in table B4. The exponential model fits the singlehood

³⁰ The transition rate to marriage is obtained after conditioning out unobserved heterogeneity in types.

spells of black men quite well (cols. 4–6). The shape parameter in singlehood spells is fairly close to one, and the asymptotic confidence interval straddles 1.0 for $\alpha < 1$. However, the model has difficulty in fitting black men's marriage spells and white men's singlehood and marriage spells data. Results for black men's marriage spells show that the standard errors of ρ are enormous. These results raise suspicion of the model fitness even though the asymptotic confidence interval covers one when $\alpha = (0, 1)$, and it is right around one to two decimal points at the corner cases (cols. 4–6). The estimates for singlehood and marriage spells for white men are around 0.4 and 0.6, respectively, for $0 < \alpha$, exhibiting decreasing singlehood and marriage hazards (cols. 1–3). These results reflect that singlehood (marriage) tenure is negatively associated with the marriage (separation) hazard. Such duration dependence may be spurious, and unobserved heterogeneity may be required to improve the fit of the model. Another way to improve the fit to marriage spells data is to discard the assumption of the exogenous separation as the only match termination mechanism. Exogenous separation can underestimate the transition rate from marriage to separation. To obtain a reasonable estimate of the transition rate of separation, one may need to extend the model to introduce endogenous separation.

VI. Conclusion

This essay is the first step in an attempt to examine empirical issues involved in structural estimation of a marriage model. The approach requires estimation of agents' types and their reservation types as a means of determining the deeper behavioral parameters. Classification errors will affect these estimates. My approach in this essay retains the behavioral matching model while adding classification errors. An application to data from the PSID shows that the model can be interpreted straightforwardly. Results reject the use of a single trait in determining agents' types and favor the use of a mixture of measured traits and classification errors. White men are found to be more impatient and their marriage market more flexible than black men. Despite that wage is found to be a more desirable marriageable trait than education for white men, its desirability exhibits a U-shaped profile with age. Older white men have a less efficient marriage market and a higher chance of being classified incorrectly. Marriage market efficiency and classification errors indicate no tractable effects of age for black men. These results reflect the improvement in understanding that who marries whom is gained by adopting structural estimation.

The advantage of the proposed method to account for classification errors is that it is flexible. Although this essay provides an intuitively appealing method to estimate a matching model, it is clear that the sep-

aration process assumed by the model is not realistic. The exogenous match destruction does not incorporate shocks that lead to destruction. More complex models to incorporate endogenous match destruction may be required for future research.

Appendix A

Derivation of the Reservation Type

From equation (1), we have

$$\beta V(x_i) = x_i + \lambda \sum_{j \in A_i} [W(x_i, x_j) - V(x_i)] \gamma_j. \quad (\text{A1})$$

Substitute equation (2) into equation (A1) to obtain

$$\beta V(x_i) = x_i + \lambda \sum_{j \in A_i} \left[\frac{x_i x_j}{2(\beta + \delta)} - \frac{\beta V(x_i)}{(\beta + \delta)} \right] \gamma_j. \quad (\text{A2})$$

The optimal policy gives $x_i R / 2(\beta + \delta) + \delta V(x_i) / (\beta + \delta) = V(x_i)$, which can be simplified to obtain

$$\frac{x_i R_i}{2} = \beta V(x_i). \quad (\text{A3})$$

Substitute equation (A3) into equation (A2),

$$\frac{x_i R_i}{2} = x_i + \lambda \sum_{j \in A_i} \left[\frac{x_i x_j}{2(\beta + \delta)} - \frac{x_i R}{2(\beta + \delta)} \right] \gamma_j. \quad (\text{A4})$$

Equation (A4) is equivalent to equation (3): $R_i = 2 + \frac{\lambda}{\beta + \delta} \sum_{j \in A_i} (x_j - R_i) \gamma_j$.

Spousal Correlations

Table A1
IPUMS, 1960–80

	<i>Wage</i>	<i>Education</i>	<i>Age</i>
1960	.3338	.6160	.7174
1970	.3054	.6063	.7516
1980	.3090	.6277	.6950

SOURCE.—The samples are drawn from the 1960, 1970, and 1980 decennial census files. I create matched family data for noninstitutionalized employed newlyweds, who have married once with current age no more than 2 years from the first age at marriage and who have no children. Age is limited to those between 15 and 64, with 15 being the official marriageable age defined by the census.

NOTE.—IPUMS = Integrated Public Use Microdata Series.

Some care must be taken in order to obtain meaningful wage correlation between spouses. To reduce measurement errors for wages, wage is trimmed using top codes provided by the census and the minimum hourly wage documented by the Bureau of the Census. Those with wages that

do not fall within admissible ranges are treated as missing. Wages used are weekly wages, and thus the minimum hourly wage is multiplied by 35 hours to get the lower wage bound.

Appendix B

Table B1
Sample Means of the PSID, 1968–93

	Mean	SD	Min	Max
White men ($N = 301$):*				
Singlehood duration*	11.45	6.05	2	41
Marriage duration	7.88	5.72	1	25
Proportion of complete singlehood spells	.6044	...	...	...
Proportion of complete marriage spells	.1196	...	...	...
Wage	461.1	234.4	106.0	1,334.0
Education	13.1	2.03	7	18
Wife's wage	315.8	160.7	80.1	1,095.9
Wife's education	12.9	2.16	3	18
Wage correlation	.500	...	...	...
Education correlation	.427	...	...	...
Black men ($N = 133$):*				
Singlehood duration*	12.56	5.80	2	34
Marriage duration	6.90	5.41	1	21
Proportion of complete singlehood spells	.5195	...	...	...
Proportion of complete marriage spells	.0752	...	...	...
Wage	432.5	219.5	113.1	1,210.1
Education	12.4	1.96	7.32	18
Wife's wage	325.4	143.5	67.5	861.0
Wife's education	12.5	1.84	7	18
Wage correlation	.453	...	...	...
Education correlation	.389	...	...	...

NOTE.—PSID = Panel Study of Income Dynamics; SD = standard deviation; min = minimum; max = maximum; N = number.

* Denotes uncensored spells only.

Table B2
Estimates from the PSID (1968–93), White Men

Age	α	λ	δ	e_0	e_1	e_2	e_3	e_4	e_5	$\log L$
Indexed by wage:										
15–21	1	.751 (.309)	.003 (.002)	.625 (.226)	.091 (.204)	.044 (.052)	.019 (.053)	.020 (.055)	.025 (.053)	–562.27
22–30	1	.716 (.306)	.002 (.002)	.449 (.209)	.116 (.204)	.050 (.051)	.047 (.051)	.044 (.055)	.037 (.052)	–3,401.26
31–40	1	.548 (.412)	.001 (.002)	.307 (.209)	.121 (.214)	.077 (.066)	.053 (.054)	.066 (.055)	.057 (.055)	–1,175.17
41–64	1	.175 (.378)	.009 (.003)	.248 (.209)	.075 (.205)	.087 (.059)	.080 (.052)	.086 (.061)	.095 (.055)	–296.42
All	1	.740 (.200)	.002 (.001)	.449 (.118)	.117 (.046)	.063 (.021)	.041 (.020)	.036 (.020)	.035 (.020)	–5,454.00

Table B2 (Continued)

Age	α	λ	δ	e_0	e_1	e_2	e_3	e_4	e_5	$\log L$
Indexed by education:										
15–21	0	.261 (.102)	.011 (.009)	.424 (.203)	.078 (.202)	.062 (.024)	.058 (.072)	.060 (.073)	.059 (.073)	–787.43
22–30	0	.168 (.108)	.018 (.010)	.306 (.201)	.096 (.201)	.062 (.021)	.079 (.071)	.072 (.071)	.076 (.072)	–3,948.33
31–40	0	.285 (.251)	.005 (.010)	.270 (.318)	.120 (.209)	.064 (.029)	.066 (.075)	.081 (.074)	.068 (.070)	–1,214.41
41–64	0	.155 (.198)	.014 (.009)	.224 (.217)	.075 (.204)	.082 (.029)	.094 (.077)	.093 (.073)	.088 (.073)	–273.11
All	0	.156 (.068)	.015 (.001)	.333 (.093)	.098 (.043)	.054 (.004)	.074 (.021)	.068 (.021)	.078 (.028)	–5,991.60
Indexed by wage and education:										
15–21	.861 (.216)	1.257 (.507)	.005 (.004)	.700 (.313)	.100 (.217)	.009 (.071)	.013 (.086)	.022 (.087)	.011 (.086)	–504.80
22–30	.826 (.109)	.776 (.462)	.003 (.004)	.508 (.339)	.096 (.217)	.039 (.050)	.044 (.086)	.034 (.087)	.066 (.083)	–3,138.98
31–40	.470 (.241)	.736 (.583)	.002 (.004)	.383 (.314)	.134 (.219)	.057 (.064)	.046 (.087)	.042 (.086)	.058 (.085)	–1,117.37
41–64	.959 (.201)	.437 (.502)	.002 (.004)	.319 (.330)	.098 (.217)	.078 (.064)	.048 (.086)	.078 (.087)	.075 (.087)	–291.88
All	.816 (.082)	.801 (.208)	.002 (.001)	.493 (.138)	.104 (.043)	.055 (.026)	.023 (.025)	.043 (.020)	.058 (.022)	–5,184.16

NOTE.—PSID = Panel Study of Income Dynamics; $e_0 = 1 - 2e_1 - 2e_2 - 2e_3 - 2e_4 - e_5$; standard errors are in parentheses.

Table B3
Estimates from the PSID (1968–93), Black Men

Age	α	λ	δ	e_0	e_1	e_2	e_3	e_4	e_5	$\log L$
By wage:										
15–21	1	.372 (.218)	.012 (.018)	.824 (.200)	.082 (.114)	.001 (.057)	.001 (.058)	.003 (.071)	.001 (.004)	–74.12
22–30	1	.865 (.103)	.004 (.008)	.500 (.199)	.152 (.098)	.043 (.055)	.023 (.054)	.028 (.072)	.008 (.002)	–1,398.97
31–40	1	.620 (.220)	.008 (.018)	.286 (.224)	.109 (.104)	.085 (.054)	.069 (.053)	.059 (.074)	.072 (.004)	–1,025.07
41–64	1	.030 (.220)	.002 (.017)	.161 (.212)	.068 (.103)	.110 (.054)	.095 (.053)	.101 (.074)	.092 (.004)	–164.15
All	1	.317 (.080)	.004 (.001)	.376 (.091)	.103 (.047)	.057 (.023)	.052 (.024)	.065 (.026)	.007 (.000)	–2,552.22
By education:										
15–21	0	.278 (.140)	.017 (.011)	.482 (.205)	.112 (.104)	.048 (.056)	.039 (.052)	.041 (.071)	.038 (.056)	–150.53
22–30	0	.283 (.132)	.005 (.003)	.464 (.207)	.118 (.106)	.061 (.056)	.038 (.051)	.033 (.070)	.036 (.056)	–1,255.30
31–40	0	.170 (.140)	.012 (.012)	.327 (.232)	.075 (.102)	.064 (.056)	.083 (.052)	.077 (.071)	.077 (.055)	–678.48
41–64	0	.288 (.141)	.001 (.012)	.478 (.234)	.208 (.102)	.039 (.056)	.005 (.052)	.004 (.073)	.010 (.056)	–100.76
All	0	.205 (.092)	.004 (.001)	.430 (.108)	.099 (.043)	.053 (.025)	.060 (.020)	.051 (.028)	.043 (.026)	–2,215.45

Table B3 (Continued)

Age	α	λ	δ	e_0	e_1	e_2	e_3	e_4	e_5	$\log L$
By wage and education:										
15–21	.768 (.136)	.320 (.169)	.014 (.010)	.766 (.314)	.110 (.108)	.002 (.060)	.002 (.061)	.002 (.061)	.003 (.063)	–78.03
22–30	.004 (.102)	.289 (.128)	.006 (.003)	.449 (.208)	.121 (.102)	.054 (.054)	.039 (.057)	.045 (.057)	.032 (.056)	–1,258.05
31–40	.040 (.129)	.146 (.155)	.008 (.013)	.317 (.212)	.065 (.107)	.061 (.052)	.089 (.062)	.074 (.061)	.105 (.061)	–673.38
41–64	.209 (.189)	.162 (.156)	.002 (.012)	.565 (.281)	.194 (.117)	.005 (.055)	.001 (.062)	.004 (.061)	.026 (.061)	–72.56
All	.031 (.012)	.205 (.079)	.006 (.001)	.436 (.107)	.094 (.054)	.050 (.029)	.055 (.029)	.054 (.025)	.058 (.024)	–2,221.07

NOTE.—PSID = Panel Study of Income Dynamics; $e_0 = 1 - 2e_1 - 2e_2 - 2e_3 - 2e_4 - e_5$; standard errors are in parentheses.

Table B4
Specification Tests for Exponential Search Times, PSID (1968–93)

	White Men			Black Men		
	ρ	95% LL	95% UL	ρ	95% LL	95% UL
Singlehood spells:						
$\alpha = 1$	.372	.332	.412	.718	.630	.806
$\alpha = 0$	1.791	1.641	1.941	1.145	.995	1.295
$\alpha = (0, 1)$	.354	.314	.394	1.145	.995	1.295
Marriage spells:						
$\alpha = 1$	.578	.492	.664	.780	.590	.970
$\alpha = 0$	1.053	.893	1.213	.780	.590	.970
$\alpha = (0, 1)$	.578	.492	.664	.874	.658	1.090

NOTE.—PSID = Panel Study of Income Dynamics; LL = lower limit; UL = upper limit.

Appendix C

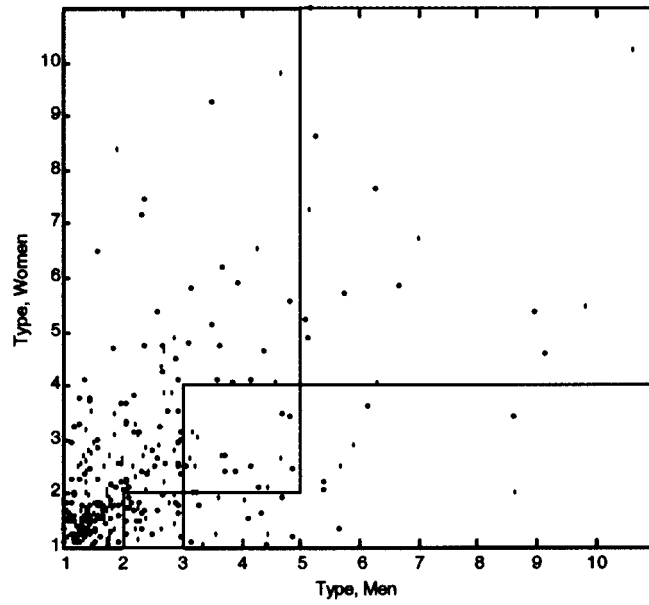


FIG. C1.—White men's acceptance sets, $\alpha = 1$

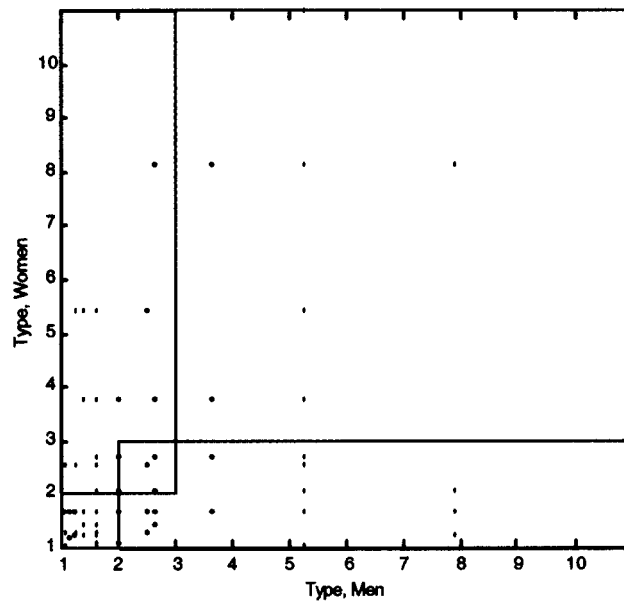


FIG. C2.—White men's acceptance sets, $\alpha = 0$

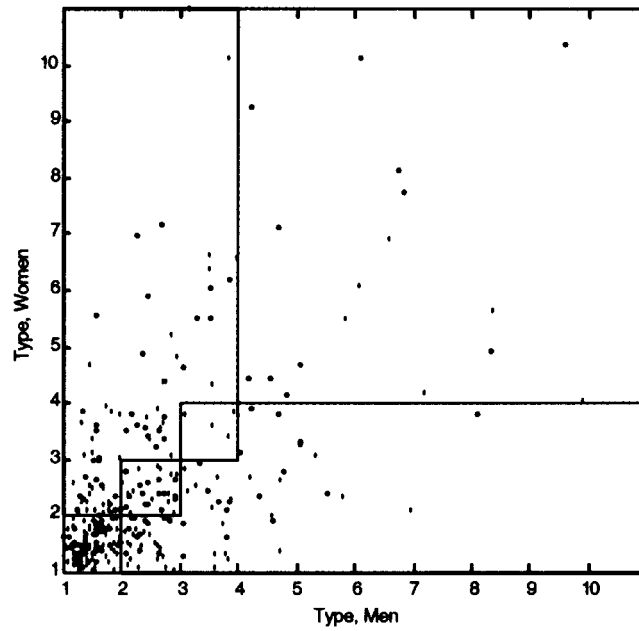


FIG. C3.—White men's acceptance sets, $\alpha = 0.816$

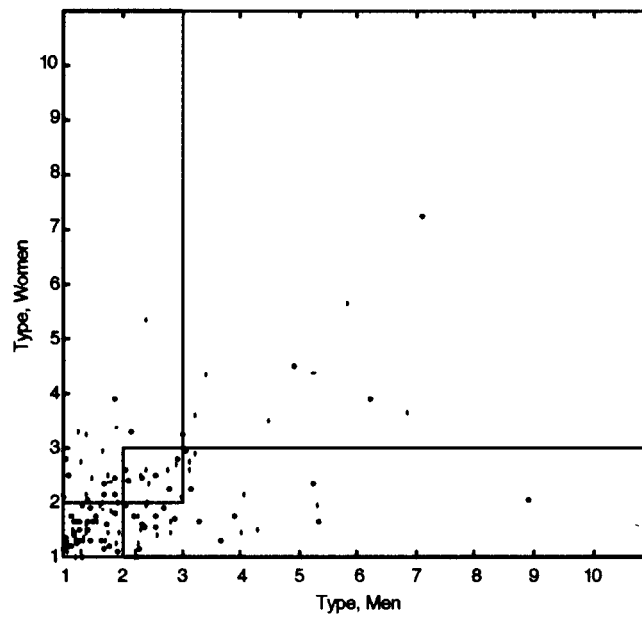


FIG. C4.—Black men's acceptance sets, $\alpha = 1$

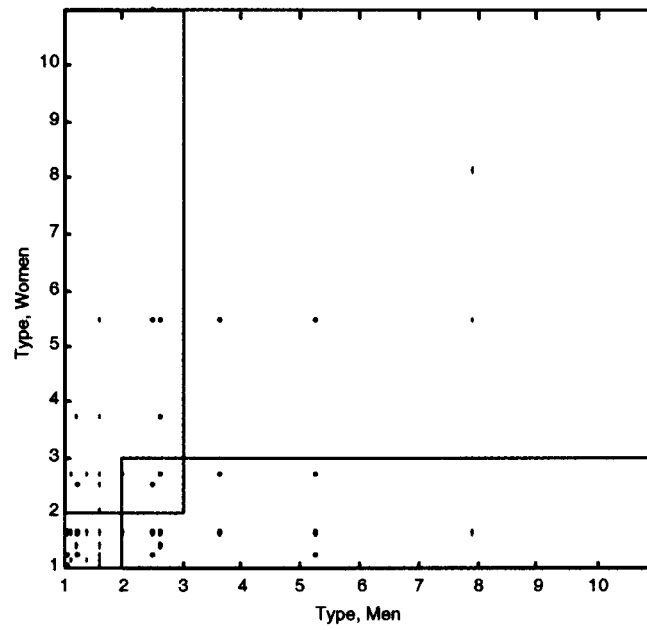


FIG. C5.—Black men's acceptance sets, $\alpha = 0$

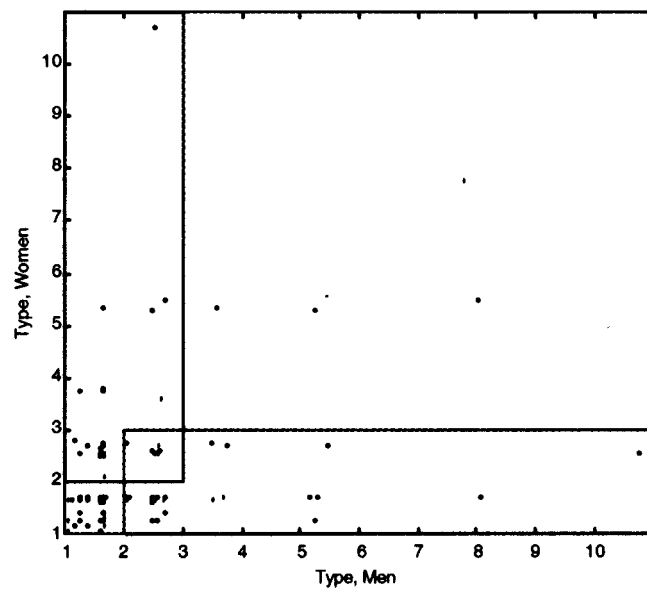


FIG. C6.—Black men's acceptance sets, $\alpha = .031$

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